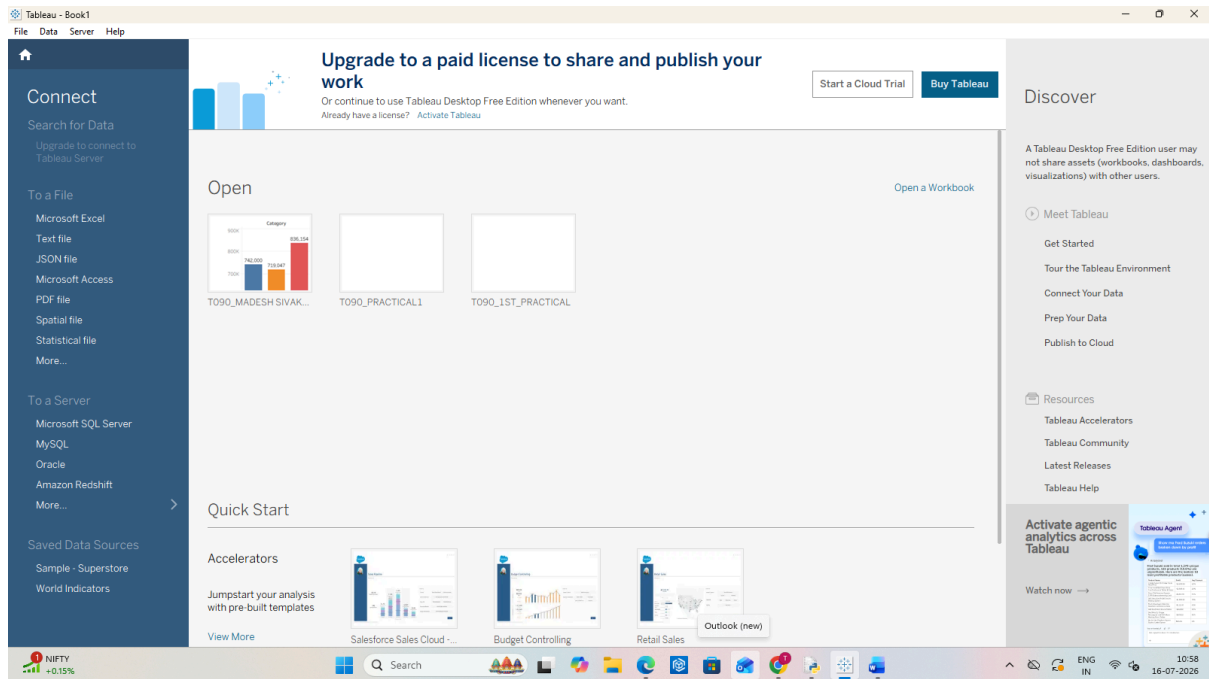


PRACTICAL NO 2

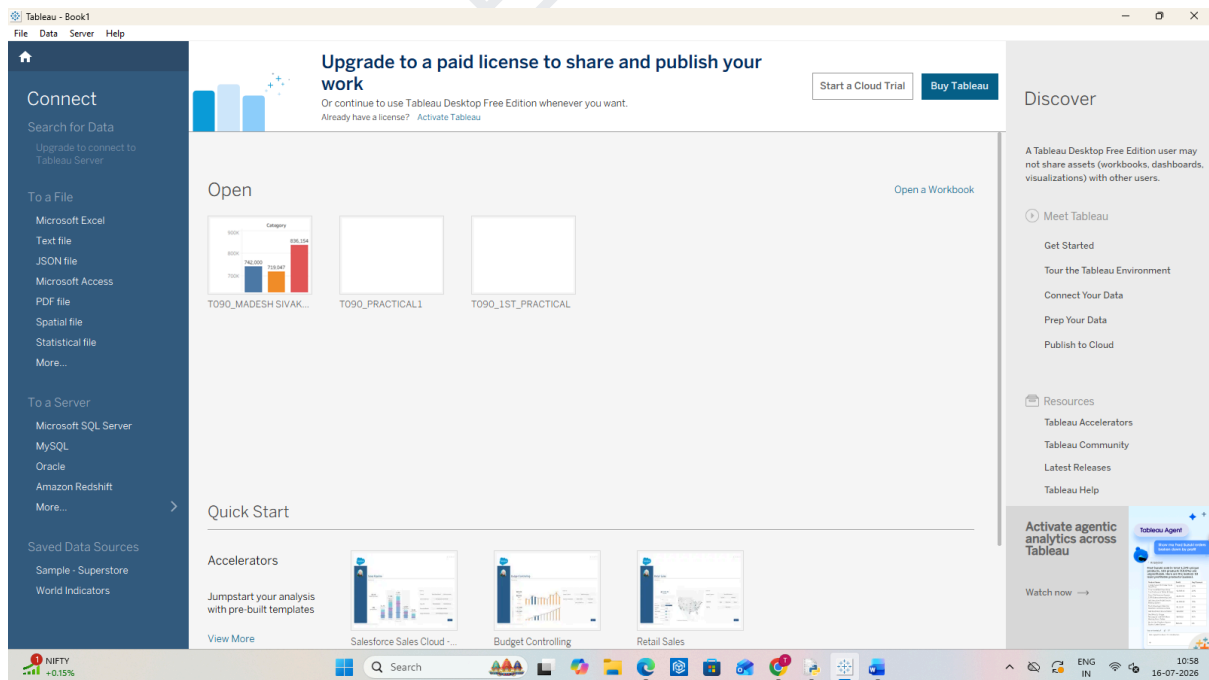
AIM: Working with Data Sources and Data Preparation in Tableau

Step A: Connecting to Different Data Sources

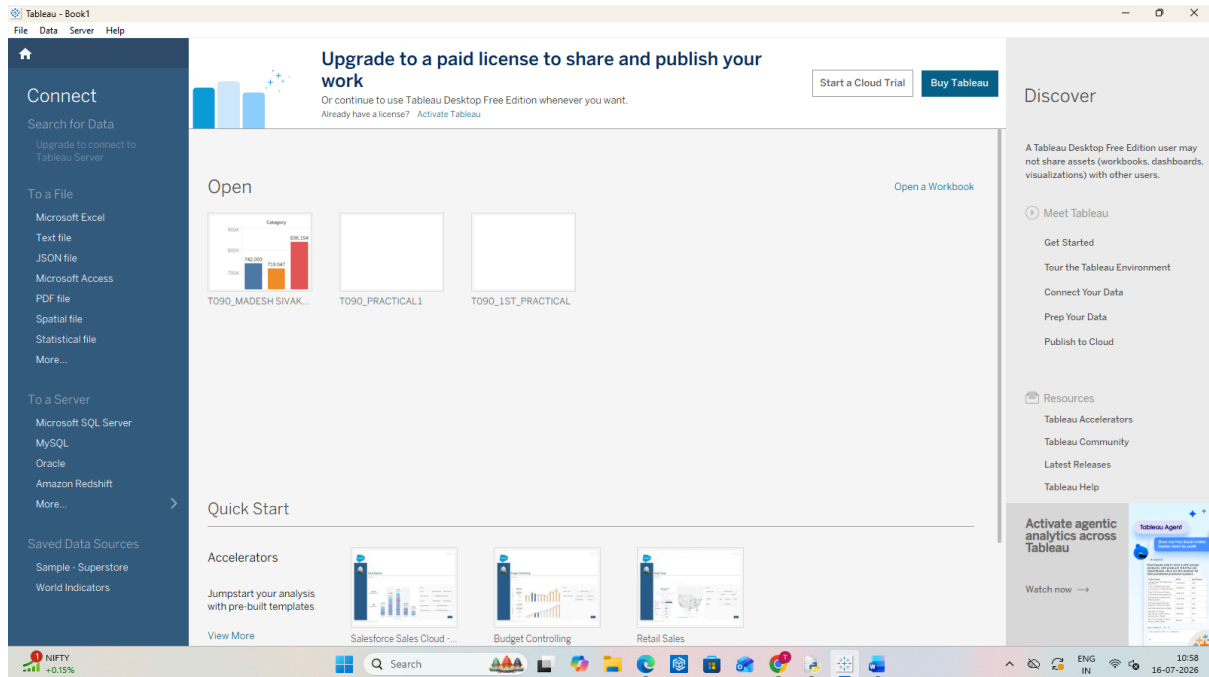
1. Launch the Tableau Desktop application.



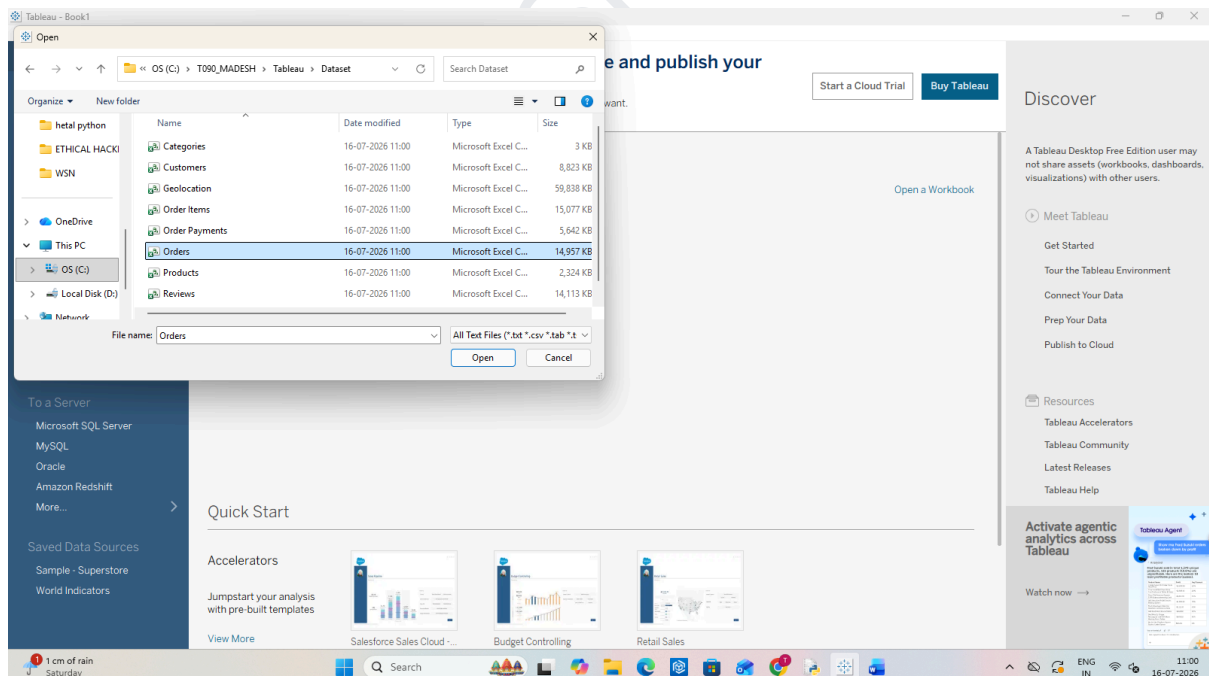
2. Locate the blue Connect control panel on the far-left section of the start page.



3. Under the To a File sub-header, click on Text File (Tableau classifies .csv spreadsheet extensions as flat text files).

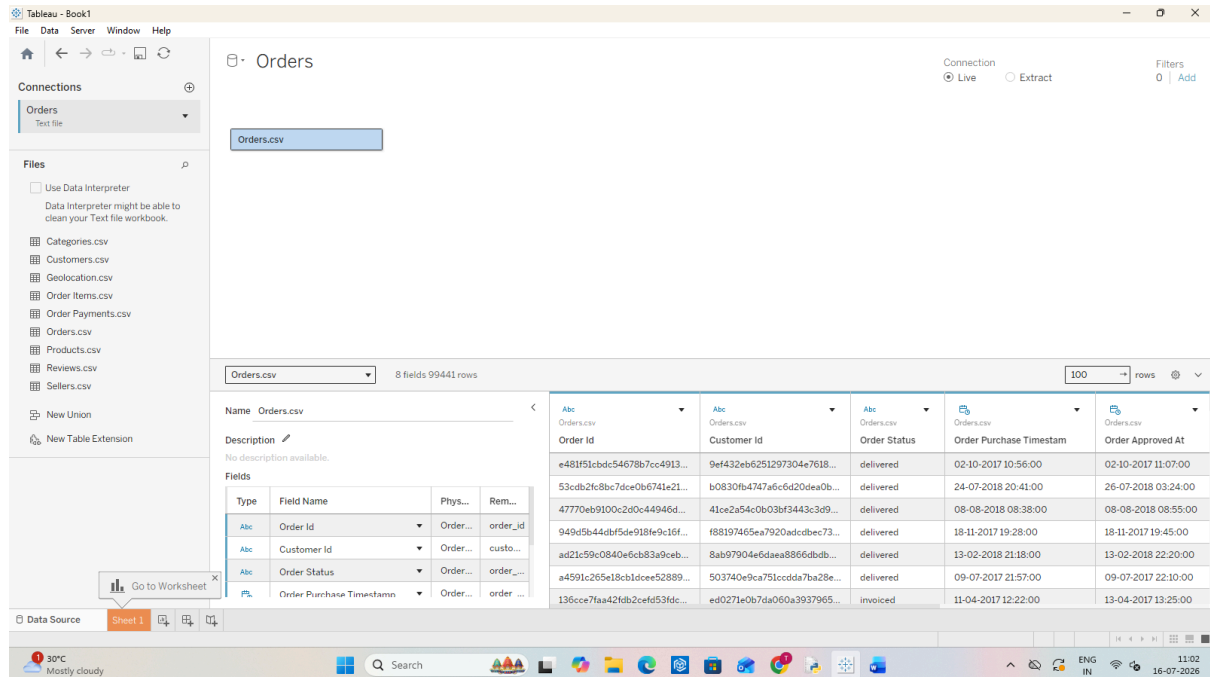


4. Navigate through your file directory, select Orders.csv, and click Open.

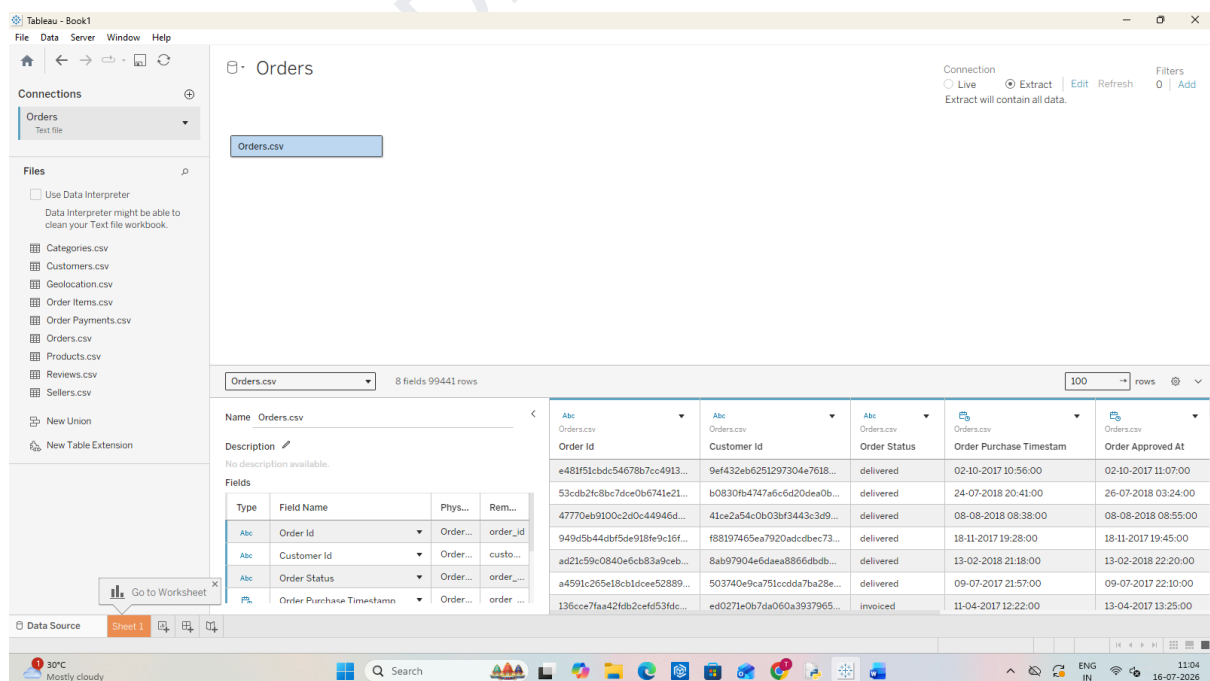


Step B: Creating Data Extracts vs. Live Connections

1. Once the dataset populates the main staging view, navigate to the top-right corner of the Data Source interface.
2. Locate the connection mode toggle section.

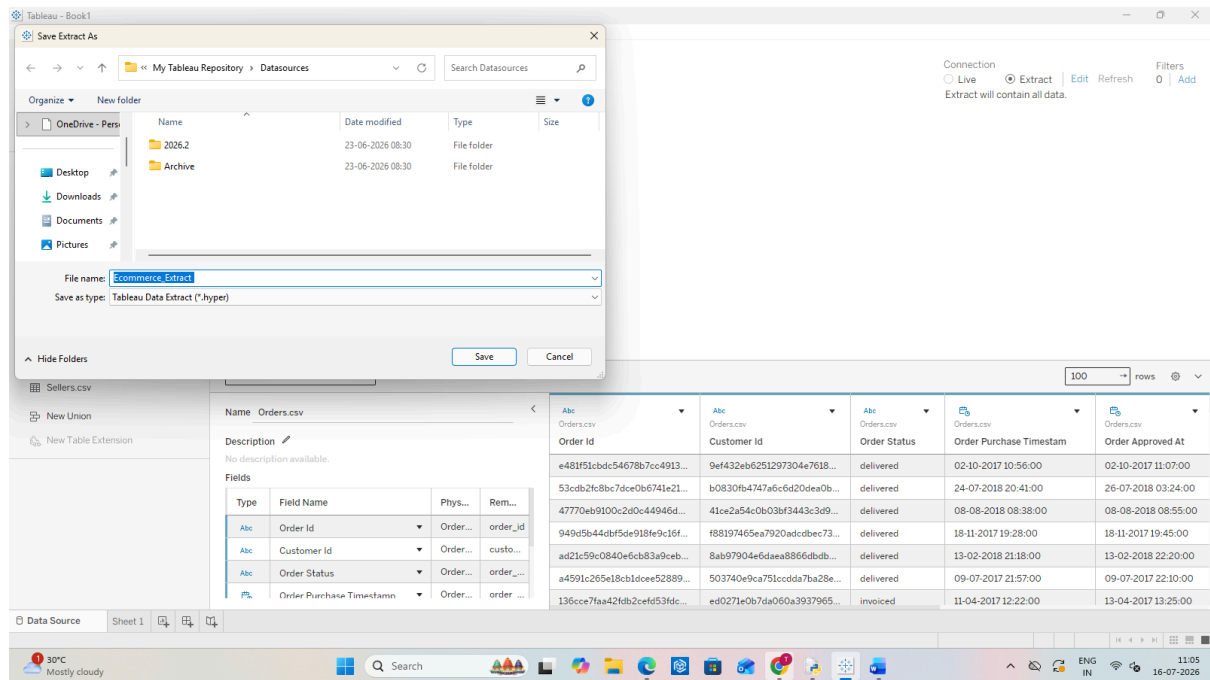


3. Switch the active selection from Live over to Extract.



4. Click on the Sheet 1 navigation tab located at the bottom-left corner of the workspace screen.

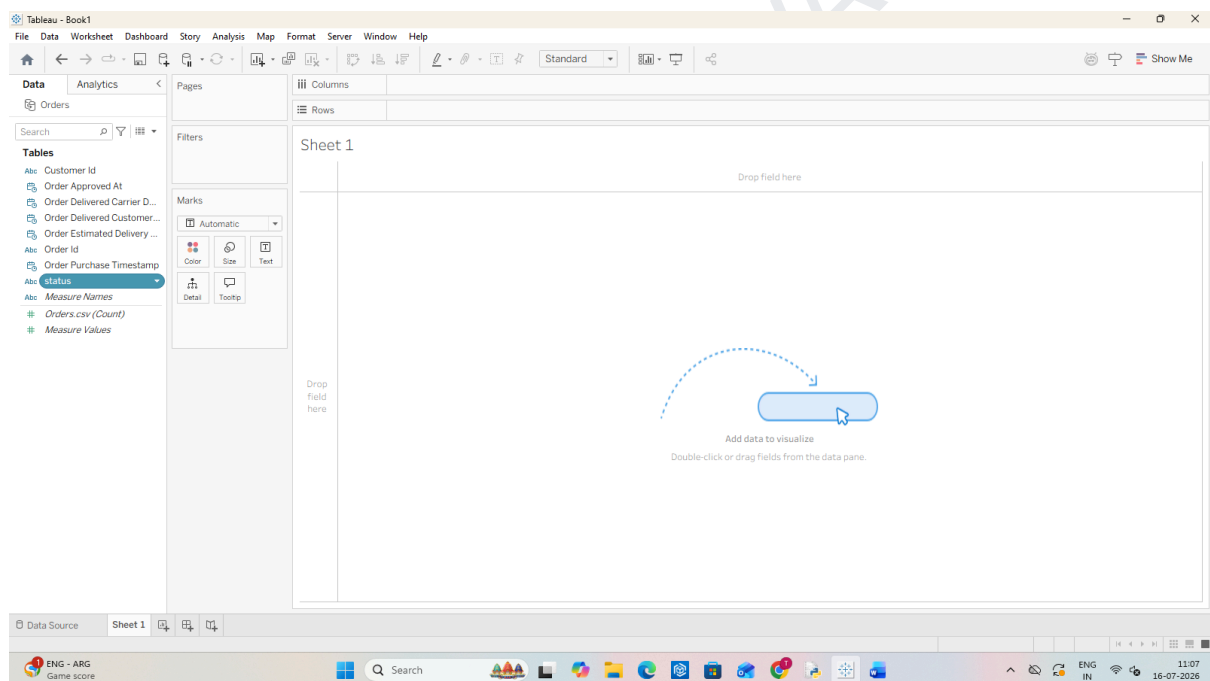
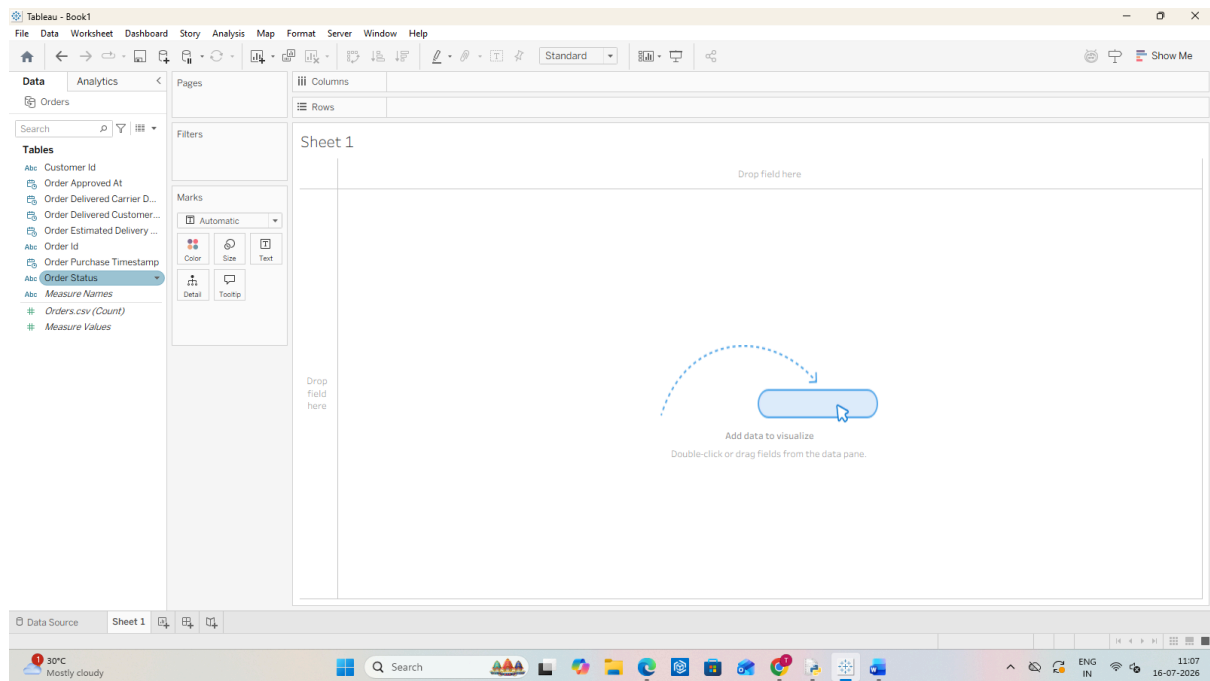
5. A system dialog save window will immediately appear. Name the local snapshot file Ecommerce_Extract and click Save.



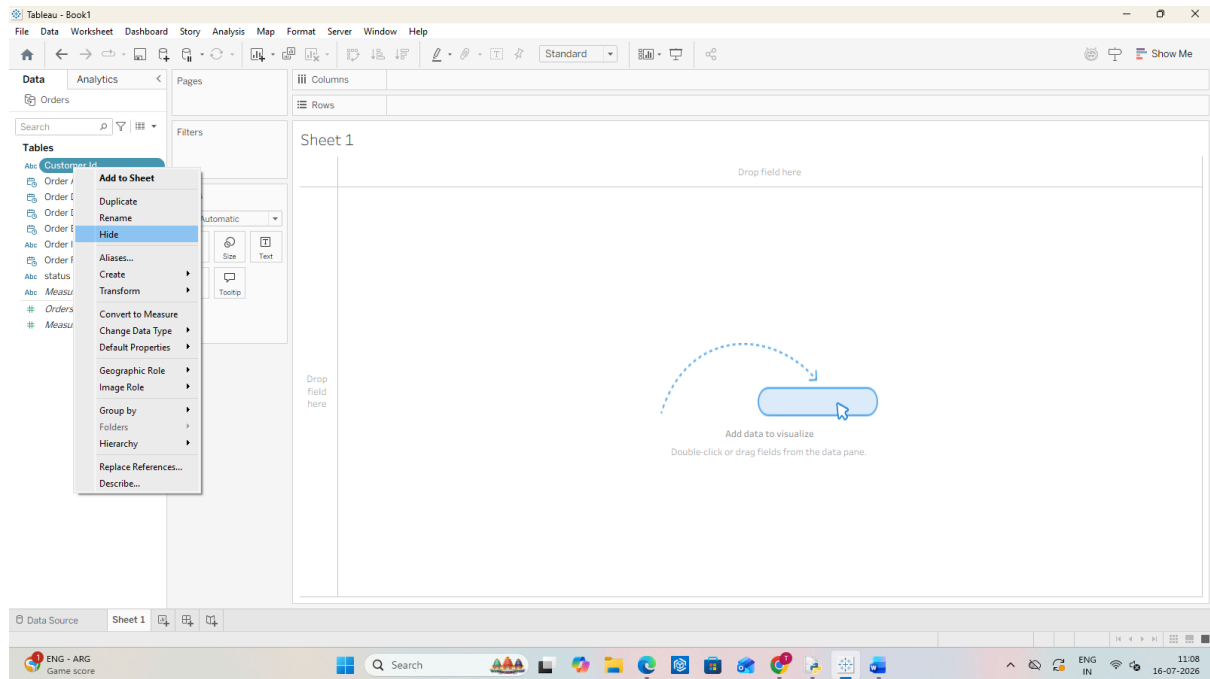
Step C: Managing Operational Metadata 1. Navigate to the blank workspace canvas on Sheet 1 and view the left-hand side Data Pane.

2. Rename a Field: Locate the dimensional attribute field labeled order_status or Order Status. Right-click the field name, select Rename, type in Status, and hit Enter.

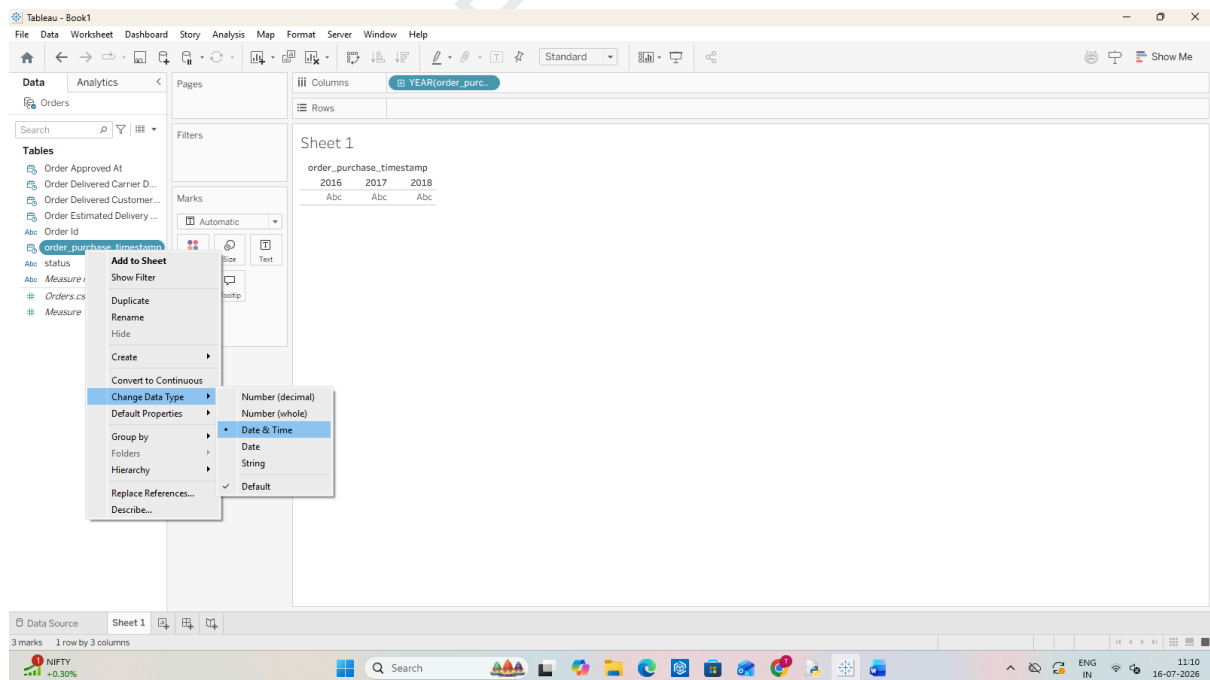
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3. Hide a Field: Locate an internal tracking string variable that will not be used in visual reporting, such as customer. Right-click the field name and select Hide.



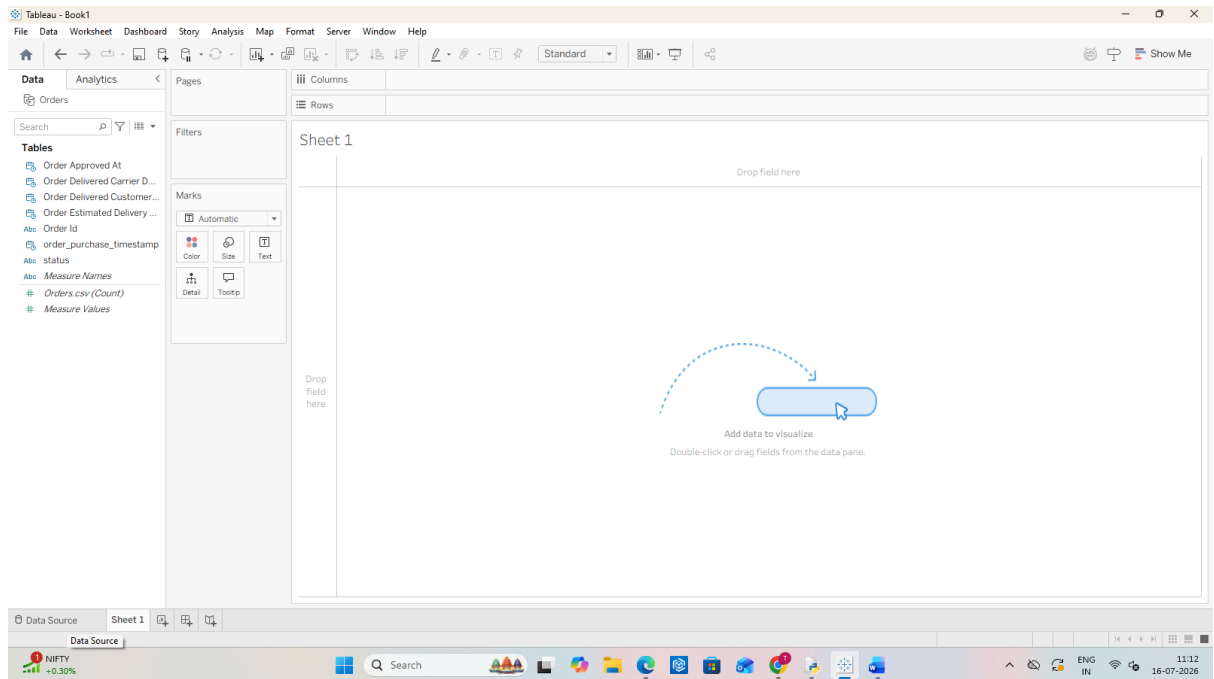
4. Change a Data Type: Locate the operational timestamp column labeled `order_purchase_timestamp`. Click directly on the data type icon (which may incorrectly display as text `Abc` or a calendar date) positioned immediately to the left of the field name. Select **Date & Time** from the context dropdown menu to force proper structural recognition (Note: It may by default selected to Date and Time).



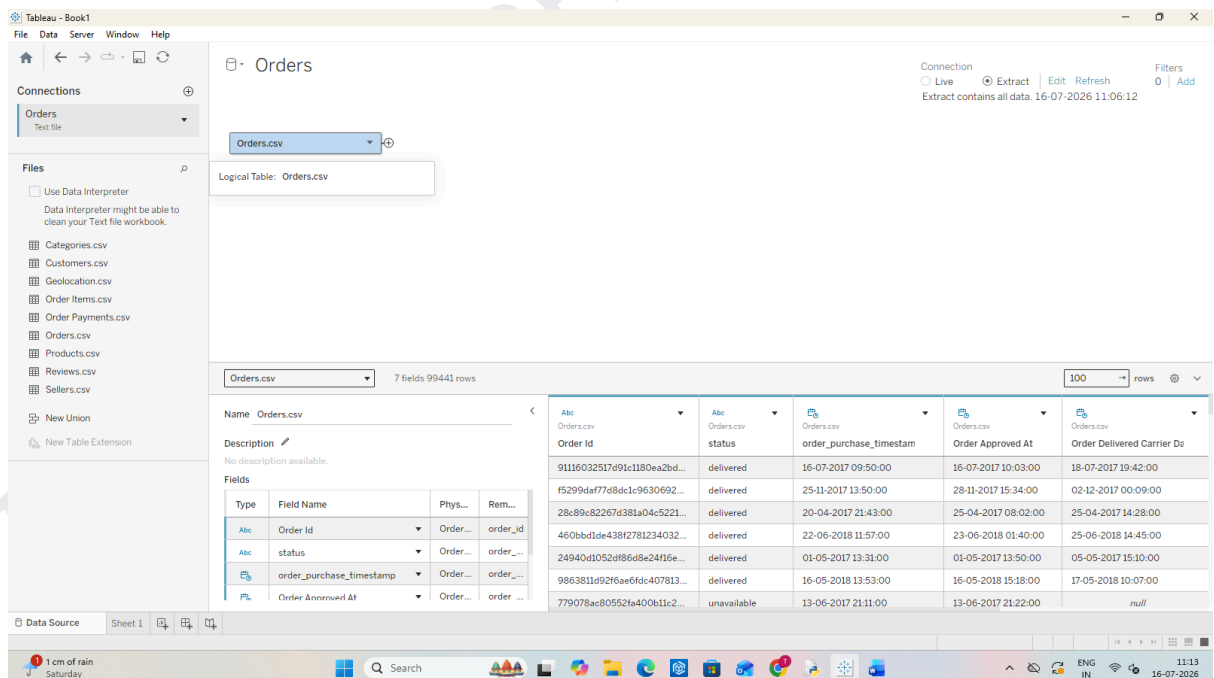
Step D: Performing Relational Joins on Multiple Tables

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1. Click the Data Source tab positioned at the bottom-left corner of the window to return to the data schema designer.



2. In the top staging canvas area, locate the rectangular Orders.csv database block.



3. Double-click directly on the Orders.csv block. This action drills down through

the logical relationship layer and enters the Physical Layer canvas space.

The screenshot shows the Tableau Desktop interface. On the left, the 'Files' pane lists several CSV files: Categories.csv, Customers.csv, Geolocation.csv, Order Items.csv, Order Payments.csv, Orders.csv, Products.csv, Reviews.csv, and Sellers.csv. The 'Orders.csv' file is selected. The main view displays the 'Orders.csv' table with 7 fields and 99,441 rows. A data preview table is shown at the bottom, listing columns: Order Id, status, order_purchase_timestamp, Order Approved At, and Order Delivered Carrier Da. The table contains several rows of data, including order IDs, status (delivered, unavailable), timestamps, and approval/delivery dates.

Order Id	status	order_purchase_timestamp	Order Approved At	Order Delivered Carrier Da
91116032517d91c1180ea2bd...	delivered	16-07-2017 09:50:00	16-07-2017 10:03:00	18-07-2017 19:42:00
f5299da77d8dc1c9630692...	delivered	25-11-2017 13:50:00	28-11-2017 15:34:00	02-12-2017 00:09:00
28c89c82267d381a04c5221...	delivered	20-04-2017 21:43:00	25-04-2017 08:02:00	25-04-2017 14:28:00
460bbd1de438f2781234032...	delivered	22-06-2018 11:57:00	23-06-2018 01:40:00	25-06-2018 14:45:00
24940d1052df86d8e24f16e...	delivered	01-05-2017 13:31:00	01-05-2017 13:50:00	05-05-2017 15:10:00
9863811d92f6ae6fd407813...	delivered	16-05-2018 13:53:00	16-05-2018 15:18:00	17-05-2018 10:07:00
779078ac80552fa400b11c2...	unavailable	13-06-2017 21:11:00	13-06-2017 21:22:00	null

4. From the left-hand files listing pane, click and drag the Order Payments.csv file, dropping it directly next to the active Orders.csv block. A traditional Venn diagram icon link will snap open between them.

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The top screenshot shows the Tableau interface with 'Orders.csv' selected. The 'Fields' pane on the left lists various tables including 'Categories.csv', 'Customers.csv', 'Geolocation.csv', 'Order Items.csv', 'Order Payments.csv', 'Orders.csv', 'Products.csv', 'Reviews.csv', and 'Sellers.csv'. The main view displays the 'Orders.csv' table with 7 fields and 99,441 rows. The fields are: Order Id, status, order_purchase_timestamp, Order Approved At, and Order Delivered Carrier Da.

The bottom screenshot shows the same Tableau interface, but now 'Order Payments.csv' is also selected, and a Venn diagram icon is visible between the two tables. The main view displays the joined data with 12 fields and 103,886 rows. The fields are: Order Id, status, order_purchase_timestamp, Order Approved At, and Order Delivered Carrier Da.

5. Click directly on the Venn diagram icon to open the Join configuration properties box.

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Tableau - Book1

File Data Server Window Help

Connections

Orders

Text file

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Categories.csv

Customers.csv

Geolocation.csv

Order Items.csv

Order Payments.csv

Orders.csv

Products.csv

Reviews.csv

Sellers.csv

New Union

New Table Extension

Orders

Orders.csv is made of 2 tables.

Join

Inner Left Right Full Outer

Data Source Order Payments.csv

Order Id = order id (Order ...)

Add new join clause

Orders.csv 12 fields 10386 rows

100 rows

Name Orders.csv

Description

No description available.

Fields

Type	Field Name	Phys...	Rem...
ABC	Order Id	Order...	order_id
ABC	status	Order...	order_...
ABC	order_purchase_timestamp	Order...	order_...
PL	Order Approved At	Order...	order ...

Order Id

status

order_purchase_timestamp

order_purchase_timestamp

Order Approved At

Order Id

status

order_purchase_timestamp

Order Approved At

Order Delivered Carrier Da

75729cd39e546ea4083ab7f...

1d2a2c1119967c45184c9f96...

be96af68ea7ac449b7173d0...

732c9ceb9786952e76768c9f...

64c8df7b4890817f4c20233...

4259e68b0021b91e05471e0...

acfc0c21ea2b051266330ab0...

delivered

delivered

delivered

delivered

delivered

delivered

delivered

09-02-2017 19:33:00

24-11-2017 19:03:00

21-01-2018 11:46:00

05-01-2018 22:10:00

09-07-2017 13:44:00

01-08-2017 16:51:00

18-04-2017 05:49:00

09-02-2017 19:43:00

24-11-2017 22:32:00

22-01-2018 13:47:00

06-01-2018 22:07:00

11-07-2017 03:55:00

01-08-2017 17:05:00

18-04-2017 06:02:00

14-02-2017 13:52:00

28-11-2017 19:42:00

24-01-2018 19:52:00

10-01-2018 17:14:00

13-07-2017 17:55:00

03-08-2017 13:04:00

24-04-2017 13:35:00

30°C Mostly cloudy

Search

ENG IN

11:16 16-07-2026

6. Under the data grid coupling properties, select order_id as the matching data key column drop-down for both the left and right tables (Note: in some case Tableau automatically selects the Order Id).

Tableau - Book1

File Data Server Window Help

Connections

Orders

Text file

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Categories.csv

Customers.csv

Geolocation.csv

Order Items.csv

Order Payments.csv

Orders.csv

Products.csv

Reviews.csv

Sellers.csv

New Union

New Table Extension

Orders

Orders.csv is made of 2 tables.

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Inner Left Right Full Outer

Data Source Order Payments.csv

Order Id = order id (Order ...)

Add new join clause

Orders.csv 12 fields 10386 rows

100 rows

Name Orders.csv

Description

No description available.

Fields

Type	Field Name	Phys...	Rem...
ABC	Order Id	Order...	order_id
ABC	status	Order...	order_...
ABC	order_purchase_timestamp	Order...	order_...
PL	Order Approved At	Order...	order ...

Order Id

status

order_purchase_timestamp

order_purchase_timestamp

Order Approved At

Order Id

status

order_purchase_timestamp

Order Approved At

Order Delivered Carrier Da

75729cd39e546ea4083ab7f...

1d2a2c1119967c45184c9f96...

be96af68ea7ac449b7173d0...

732c9ceb9786952e76768c9f...

64c8df7b4890817f4c20233...

4259e68b0021b91e05471e0...

acfc0c21ea2b051266330ab0...

delivered

delivered

delivered

delivered

delivered

delivered

delivered

09-02-2017 19:33:00

24-11-2017 19:03:00

21-01-2018 11:46:00

05-01-2018 22:10:00

09-07-2017 13:44:00

01-08-2017 16:51:00

18-04-2017 05:49:00

09-02-2017 19:43:00

24-11-2017 22:32:00

22-01-2018 13:47:00

06-01-2018 22:07:00

11-07-2017 03:55:00

01-08-2017 17:05:00

18-04-2017 06:02:00

14-02-2017 13:52:00

28-11-2017 19:42:00

24-01-2018 19:52:00

10-01-2018 17:14:00

13-07-2017 17:55:00

03-08-2017 13:04:00

24-04-2017 13:35:00

30°C Mostly cloudy

Search

ENG IN

11:17 16-07-2026

7. Toggle Join Configurations: o Click Inner: Observe the underlying data preview window at the bottom. The rows instantly filter to show only records that contain both an order entry and a corresponding transaction payment receipt.

Tableau - Book1

Connections: Orders (Text file)

Files: Categories.csv, Customers.csv, Geolocation.csv, Order Items.csv, Order Payments.csv, Orders.csv, Products.csv, Reviews.csv, Sellers.csv

Orders.csv is made of 2 tables. ①

Join: Orders.csv (Data Source) and Order Payments.csv (Order Payments.csv)

Join Type: Left

Fields:

Type	Field Name	Phys...	Rem...
ABC	Order Id	Order...	order_id
ABC	status	Order...	order_...
ABC	order_purchase_timestamp	Order...	order_...
ABC	Order Approved At	Order...	order ...

12 fields 10386 rows

Order Id	status	order_purchase_timestamp	Order Approved At	Order Delivered Carrier De
75729cd39e546ea4083ab7f...	delivered	09-02-2017 19:33:00	09-02-2017 19:43:00	14-02-2017 13:52:00
1d2a2c1119967c45184c9f96...	delivered	24-11-2017 19:03:00	24-11-2017 22:32:00	28-11-2017 19:42:00
be96af68ea7ac449b7173d0...	delivered	21-01-2018 11:46:00	22-01-2018 13:47:00	24-01-2018 19:52:00
732c9ceb9786952e76768c9f...	delivered	05-01-2018 22:10:00	06-01-2018 22:07:00	10-01-2018 17:14:00
64c8df7b4890817dc202333...	delivered	09-07-2017 13:44:00	11-07-2017 03:55:00	13-07-2017 17:55:00
4259e68b0021b91e05471e0...	delivered	01-08-2017 16:51:00	01-08-2017 17:05:00	03-08-2017 13:04:00
acfc0c2fea2b051266330ab0...	delivered	18-04-2017 05:49:00	18-04-2017 06:02:00	24-04-2017 13:35:00

o Click Left: Notice the view updates to safely preserve 100% of the root rows coming from Orders.csv, leaving payment parameter cells completely empty if an order has no matching payment record.

Tableau - Book1

Connections: Orders (Text file)

Files: Categories.csv, Customers.csv, Geolocation.csv, Order Items.csv, Order Payments.csv, Orders.csv, Products.csv, Reviews.csv, Sellers.csv

Orders.csv is made of 2 tables. ①

Join: Orders.csv (Data Source) and Order Payments.csv (Order Payments.csv)

Join Type: Left

Fields:

Type	Field Name	Phys...	Rem...
ABC	Order Id	Order...	order_id
ABC	status	Order...	order_...
ABC	order_purchase_timestamp	Order...	order_...
ABC	Order Approved At	Order...	order ...

12 fields 10387 rows

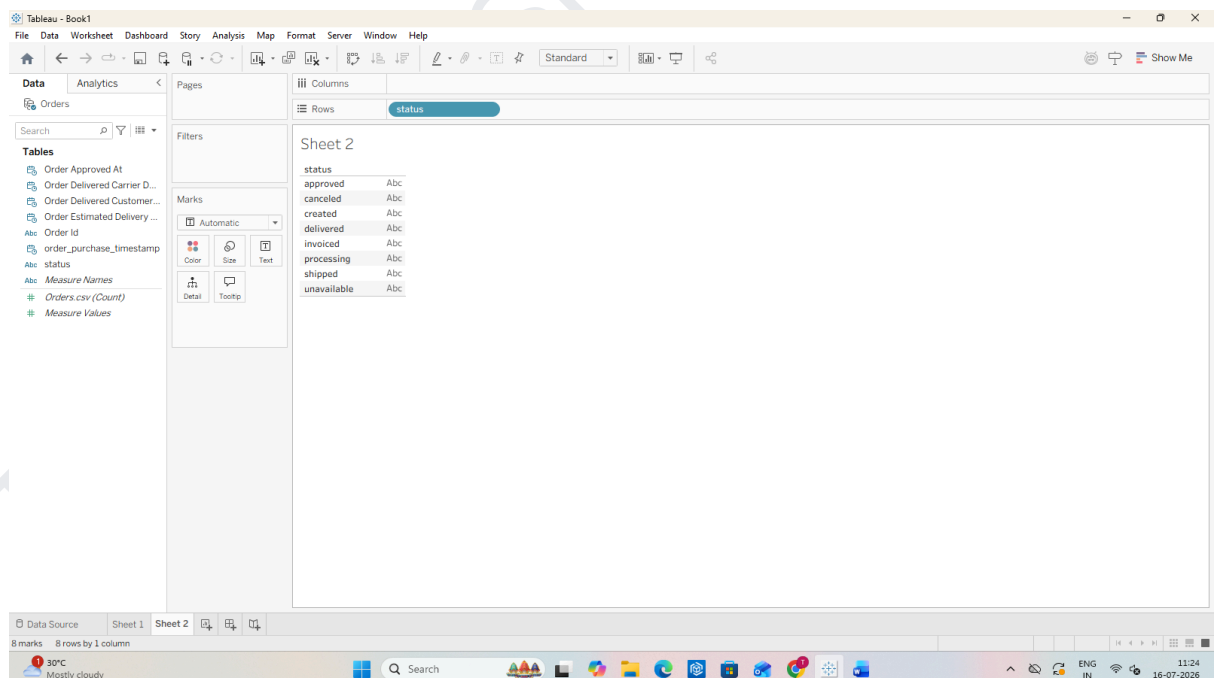
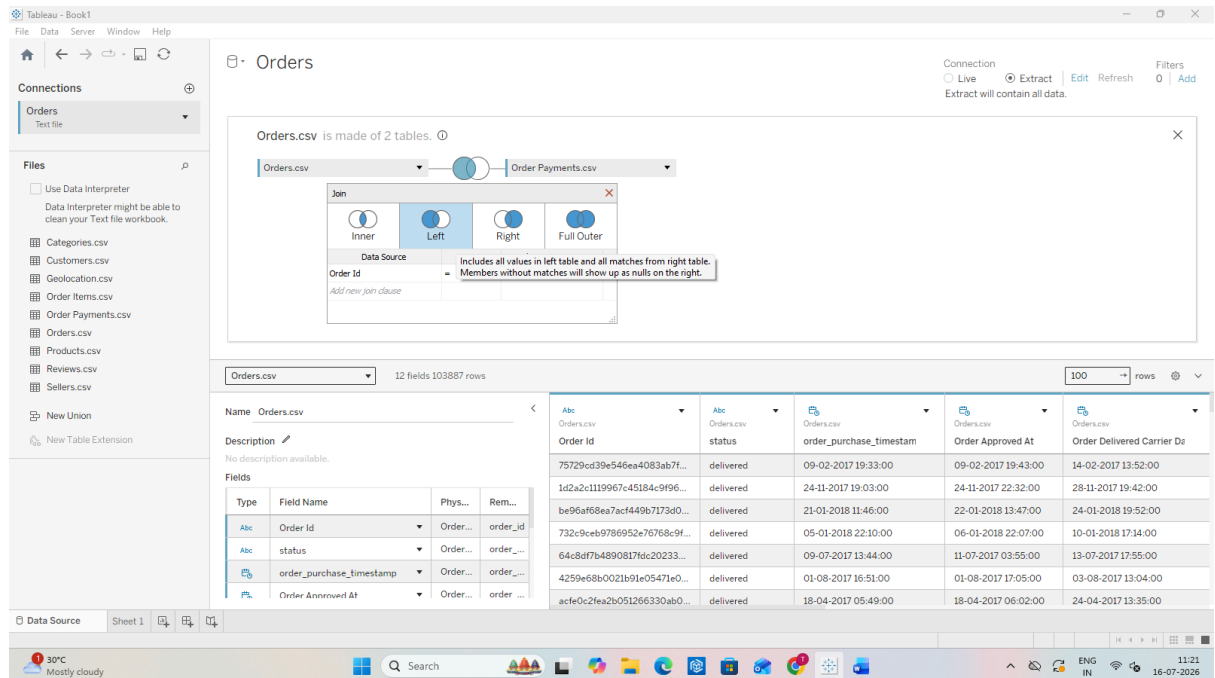
Order Id	status	order_purchase_timestamp	Order Approved At	Order Delivered Carrier De
75729cd39e546ea4083ab7f...	delivered	09-02-2017 19:33:00	09-02-2017 19:43:00	14-02-2017 13:52:00
1d2a2c1119967c45184c9f96...	delivered	24-11-2017 19:03:00	24-11-2017 22:32:00	28-11-2017 19:42:00
be96af68ea7ac449b7173d0...	delivered	21-01-2018 11:46:00	22-01-2018 13:47:00	24-01-2018 19:52:00
732c9ceb9786952e76768c9f...	delivered	05-01-2018 22:10:00	06-01-2018 22:07:00	10-01-2018 17:14:00
64c8df7b4890817dc202333...	delivered	09-07-2017 13:44:00	11-07-2017 03:55:00	13-07-2017 17:55:00
4259e68b0021b91e05471e0...	delivered	01-08-2017 16:51:00	01-08-2017 17:05:00	03-08-2017 13:04:00
acfc0c2fea2b051266330ab0...	delivered	18-04-2017 05:49:00	18-04-2017 06:02:00	24-04-2017 13:35:00

8. Leave the configuration set to a Left Join to preserve complete order history logs for the remainder of this exercise.

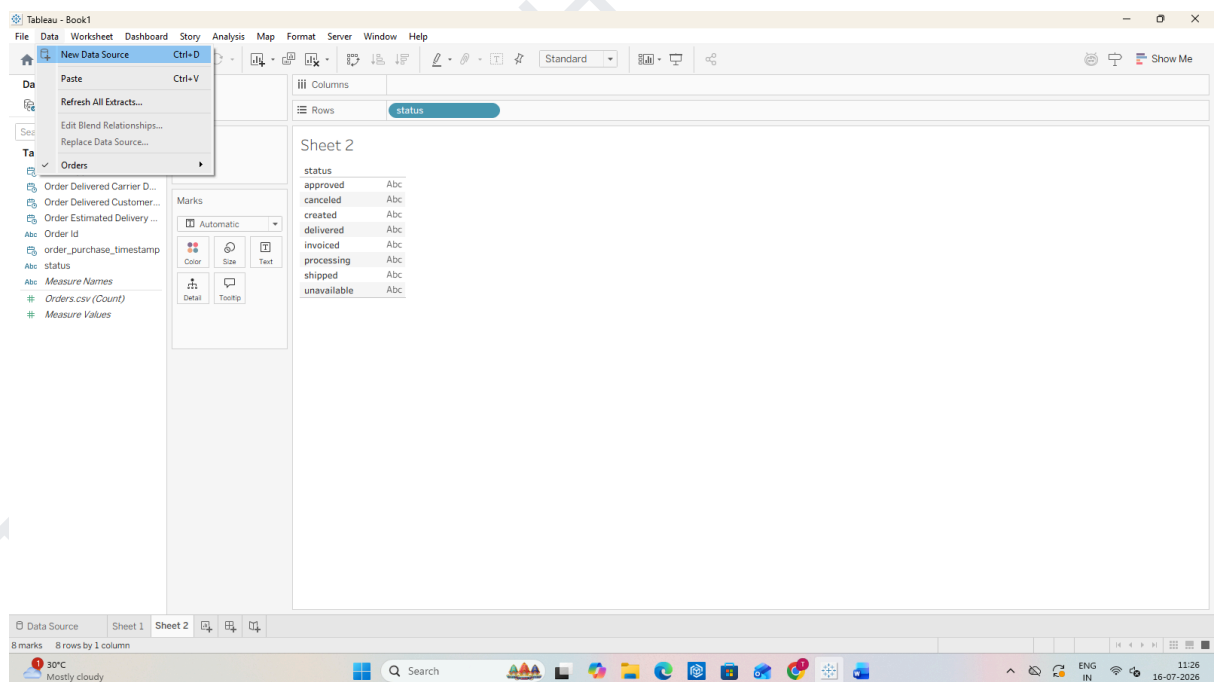
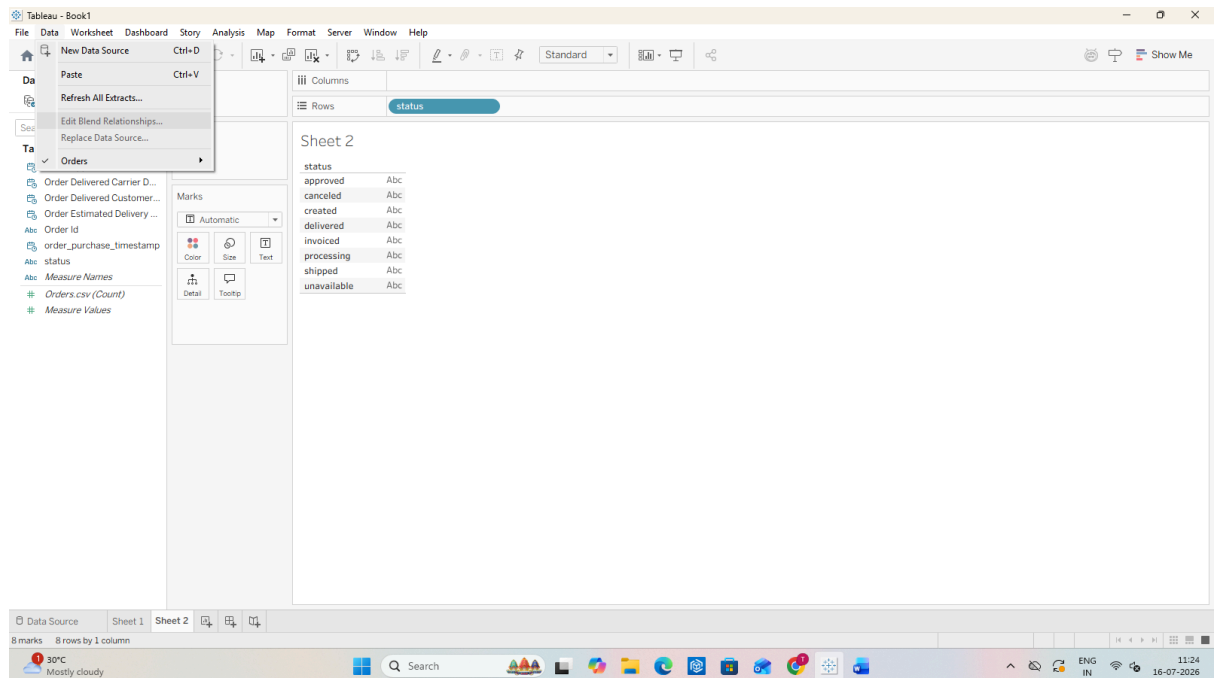
Step E: Applying Data Blending Between Independent Sources 1. Create a brand new blank workspace sheet by clicking on the New Worksheet icon (the chart

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block containing a \$\$ symbol) positioned at the bottom navigation tab area. This opens Sheet 2. 2. Ensure your joined Orders dataset connection is highlighted at the top left of the Data Pane. Drag the newly modified Status dimension field and drop it onto the Rows shelf. A blue checkmark will appear on the source, designating it as the Primary data pipeline.

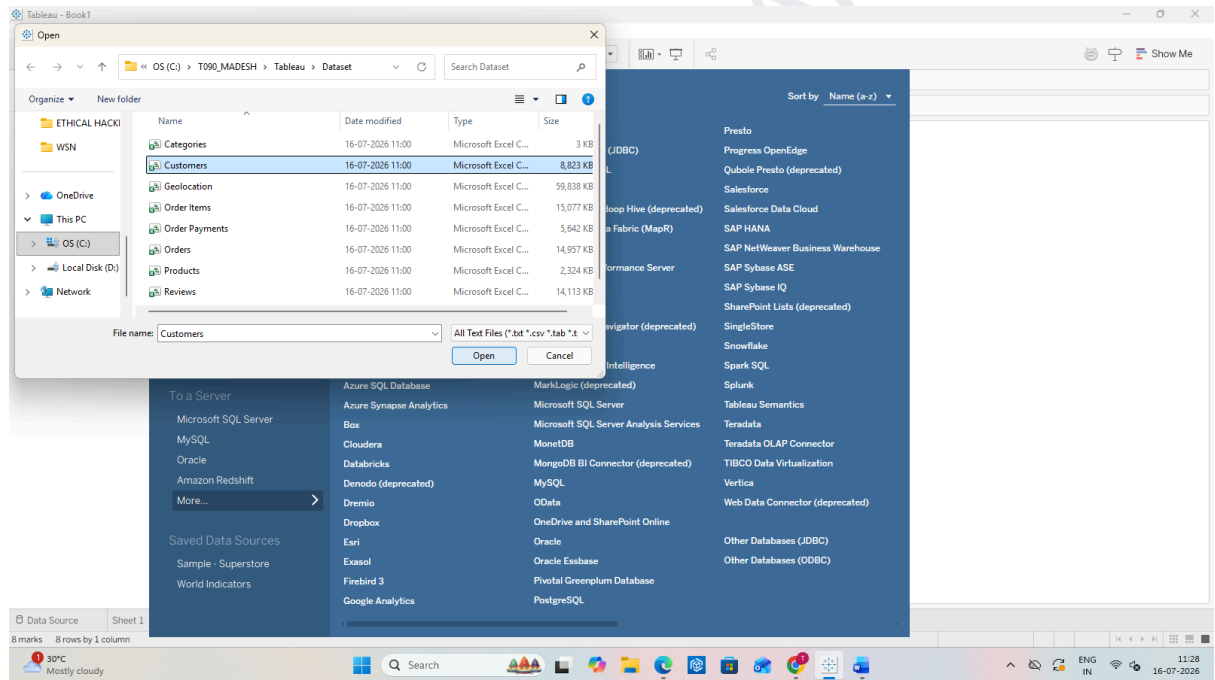
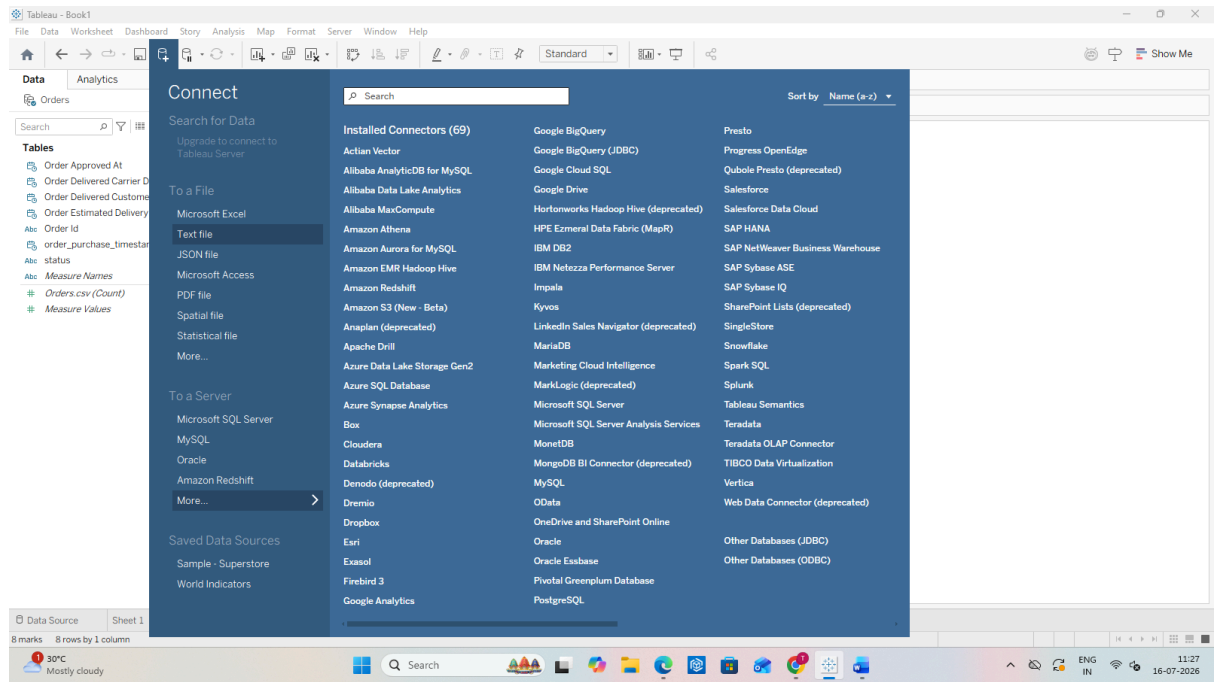


3. Move your mouse to the top application menu bar and click Data > Edit Blend Relationships. If the Option is grayed out as shown in the snapshot, click the New Data Source Option



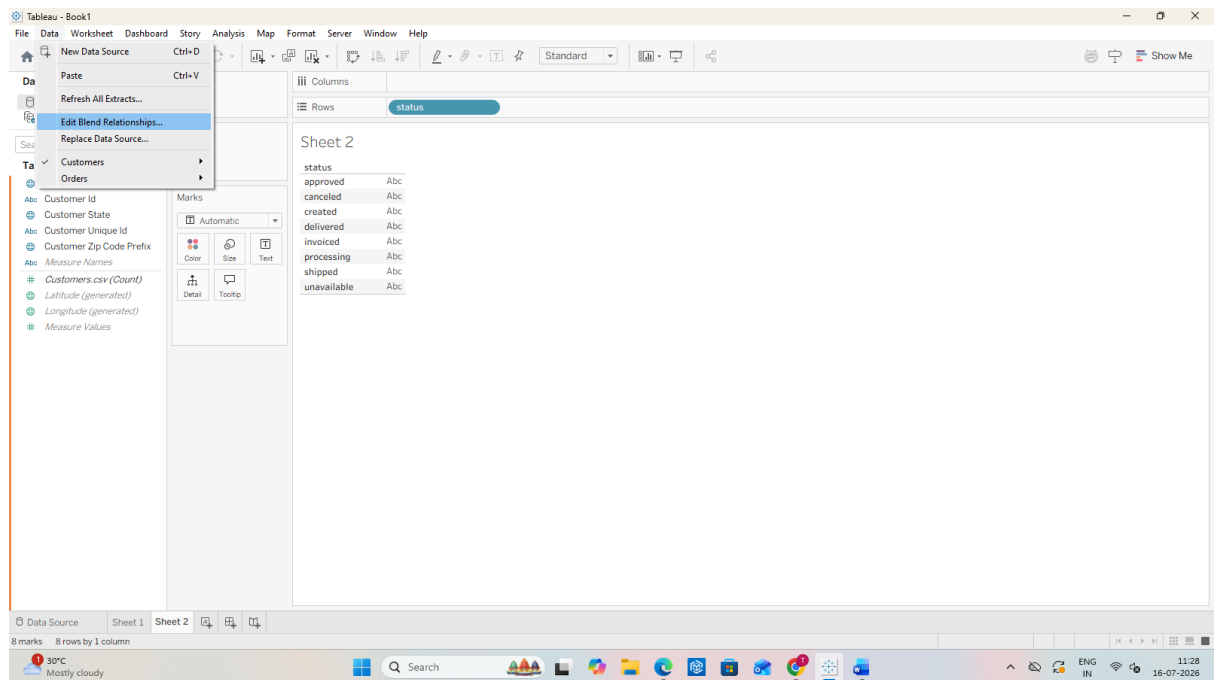
After Clicking new data source you are directed to blue menu window select text file to add the Customers.csv file.

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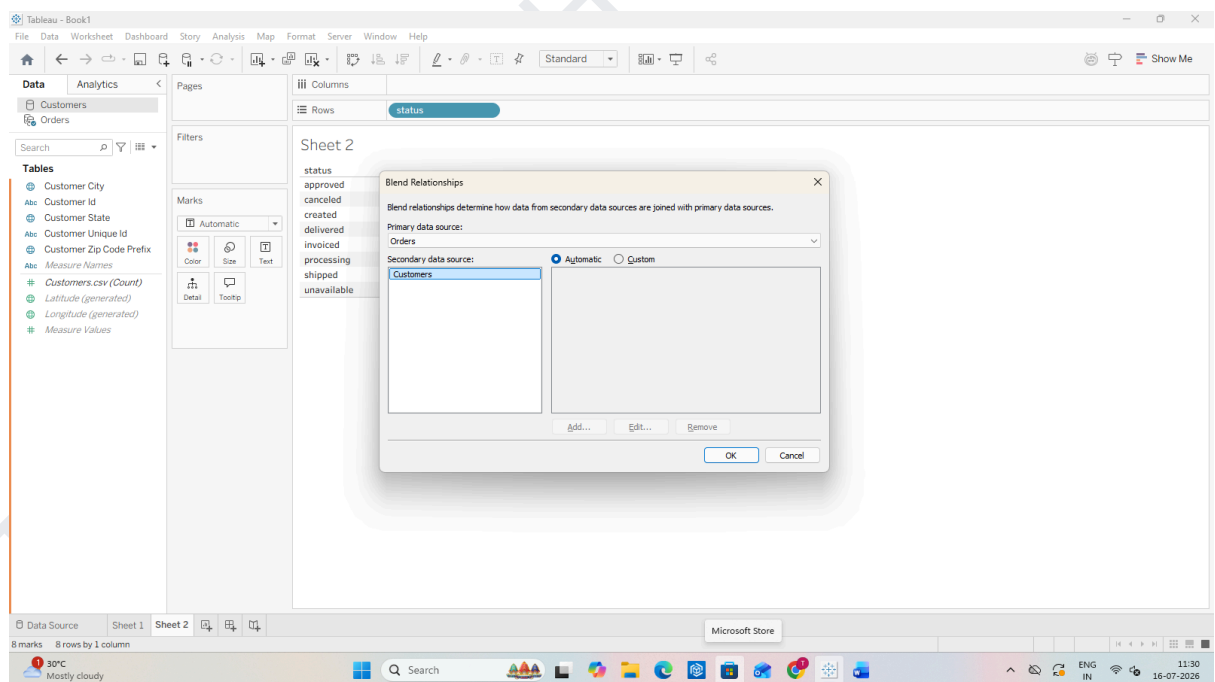


After adding the CSV file get back to sheet 2 and recheck the Edit Blend

Relationship Option and click on it

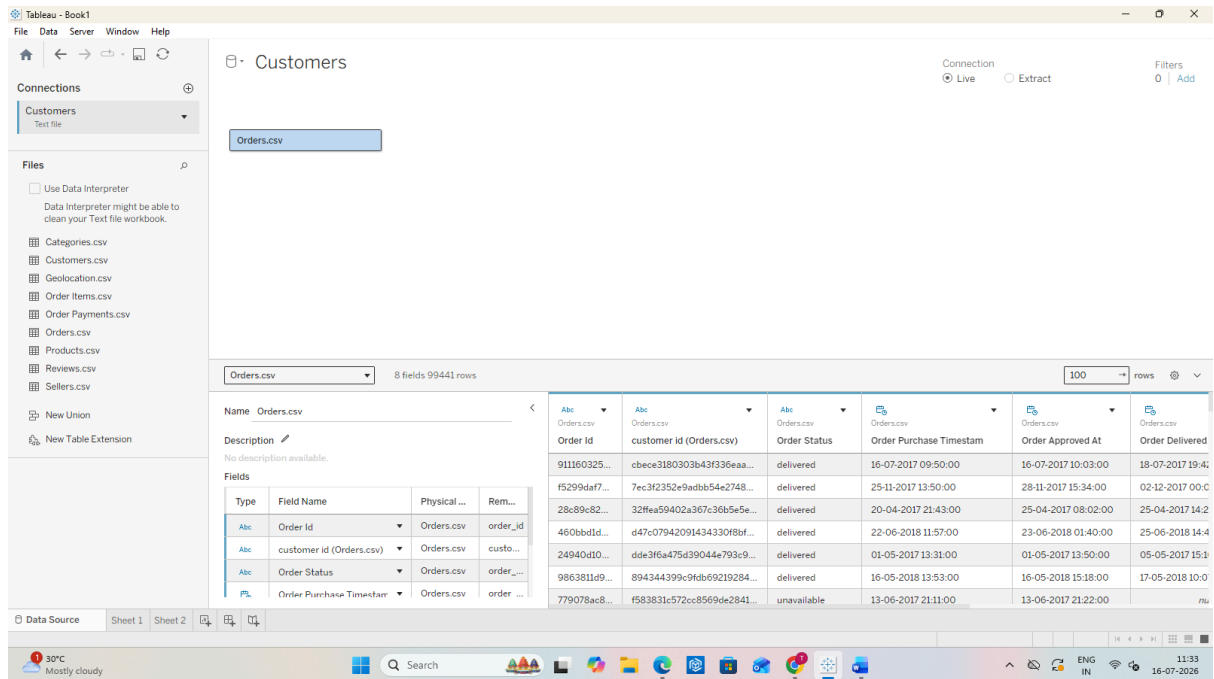


5. Switch the selection toggle switch from Automatic over to Custom, click the Add... button on the right, pair customer_id from the left dataset column to customer_id in the right column, and click OK.

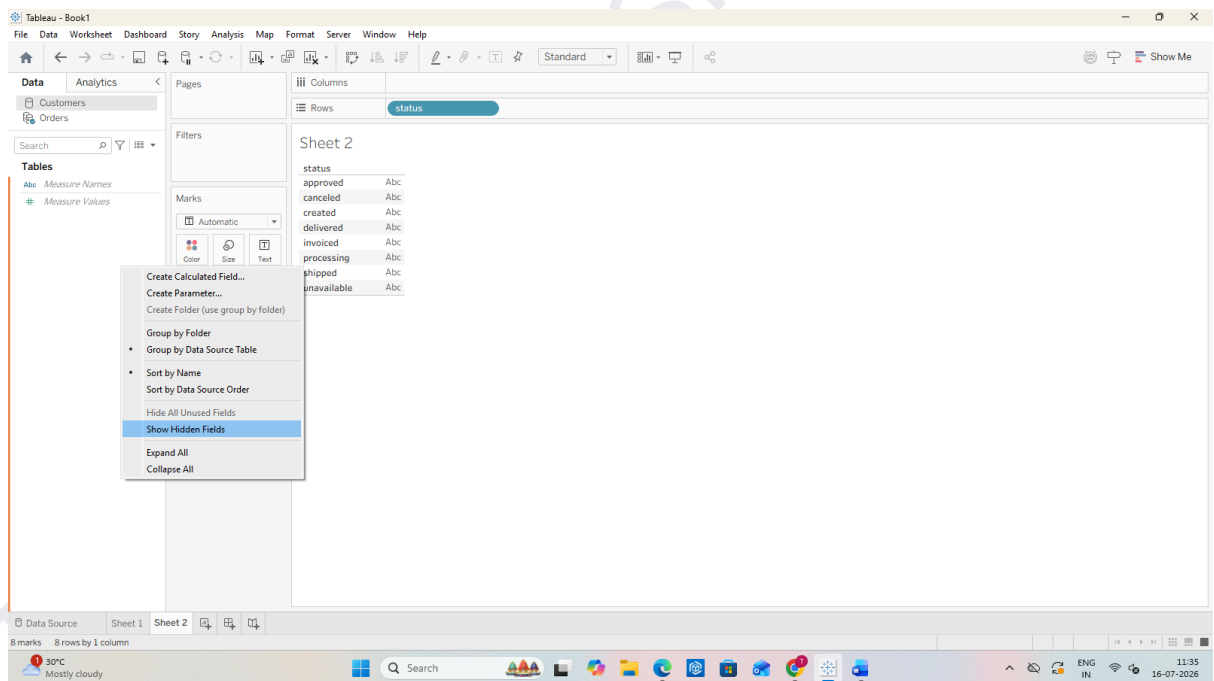


If the Customer Id is not visible in primary data its not an issue since we hid it earlier at Step C Now Navigate Back to data source and change the connection to "LIVE" from "EXTRACT"

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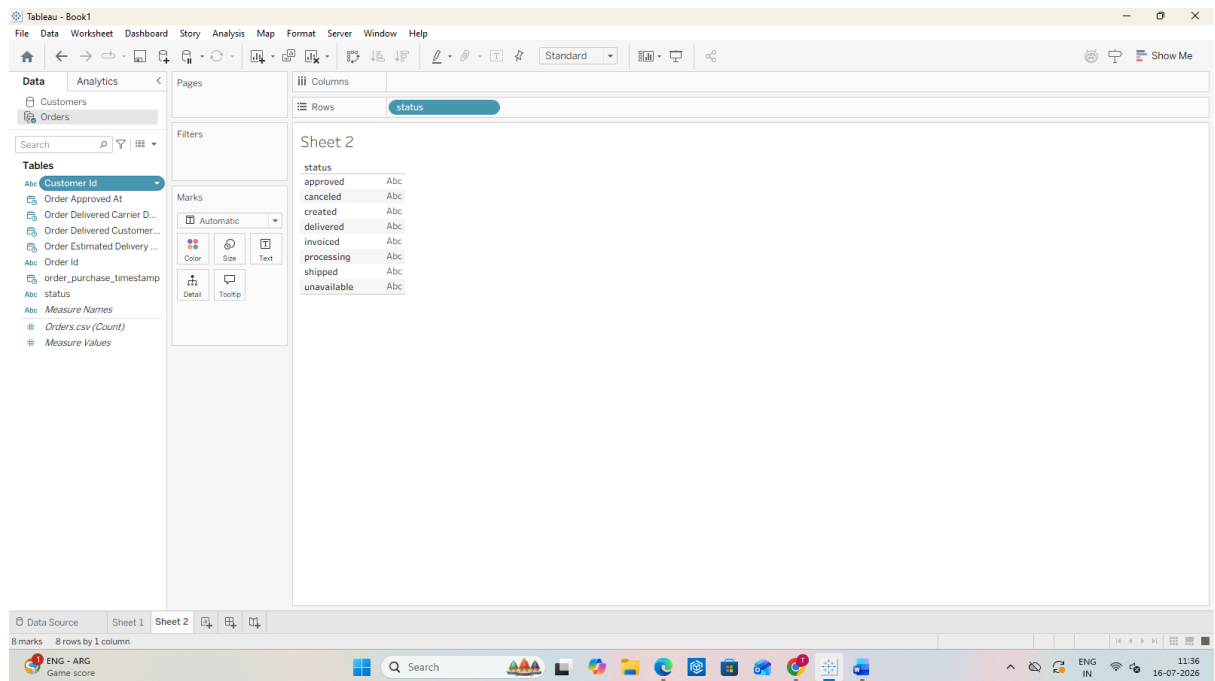


Now Go Back to any to the Sheet 2 and right click on the Data Pane's empty space and click on "Show Hidden Fields"

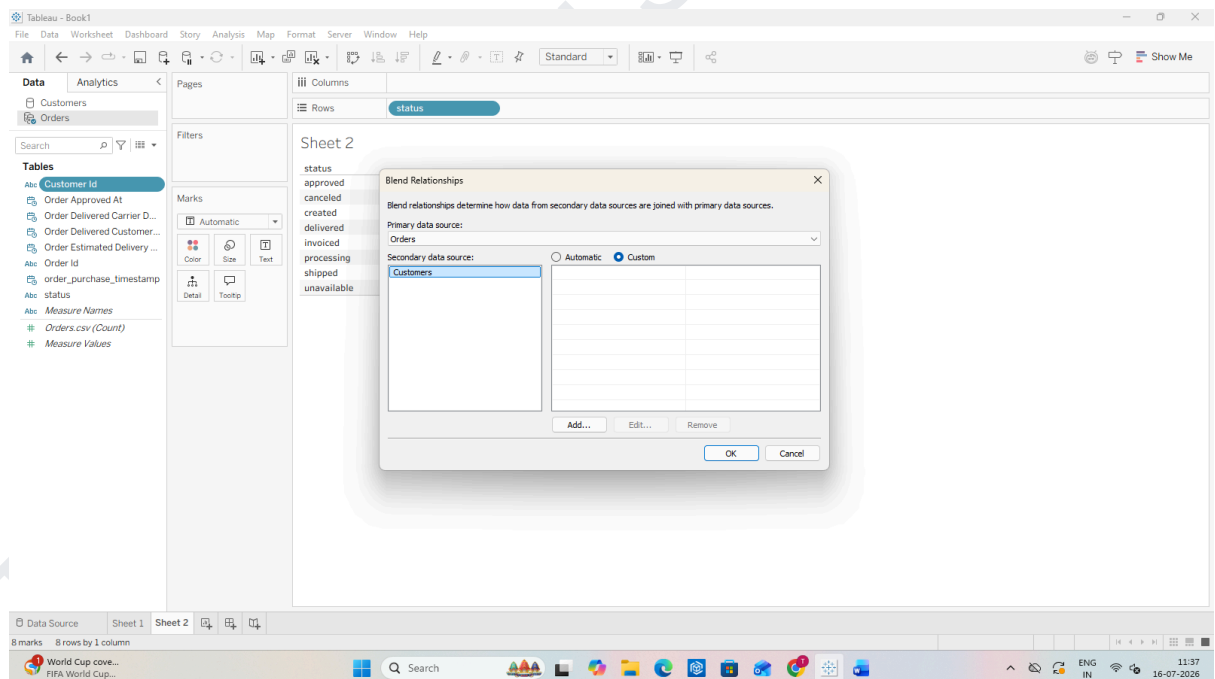


After Clicking it the Customer Id comes back into the data pane and make sure you click the "eye button" next to it marked in bellow provided snapshot

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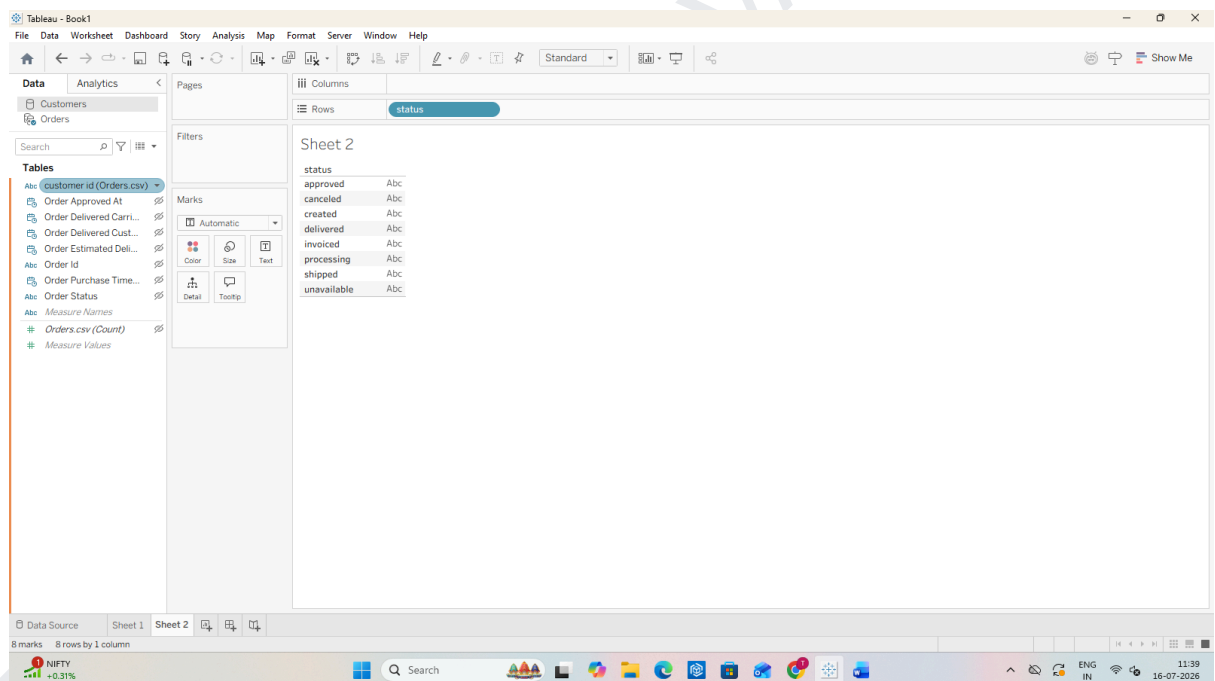
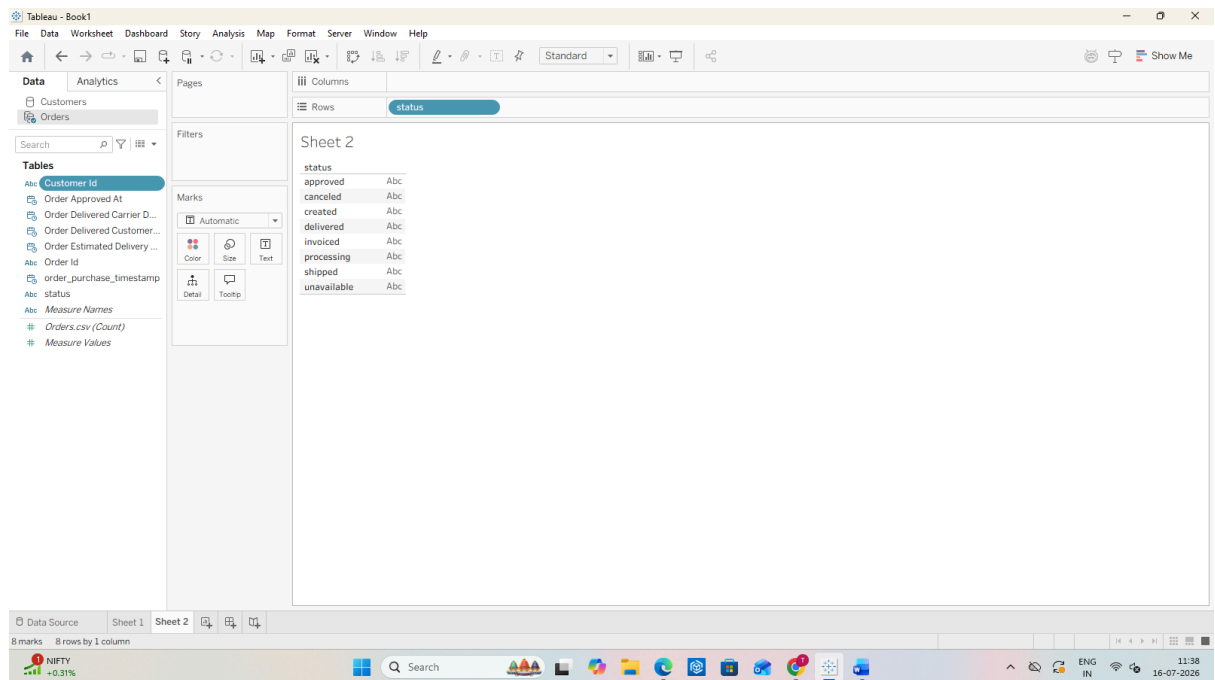


Now go back to Edit Blend Relationship option and after clicking it you may see both Customer Id in Left and Right, After that click on Custom it changes from automatic to custom and click Ok



6. Now, go back to the top-left section of your Data Pane and click directly on the separate Customers dataset tracking label. An orange chain link icon will now

appear locked together immediately next to the customer_id field.



7. Click and drag the customer_state dimension variable out from this secondary dataset list and drop it onto the Columns shelf.